

VLANs & Bonding Cheat Sheet

VLANs split one cable into several separate networks. Bonding joins several cables into one link for failover or speed.

What a VLAN tag is

```
switch port 7 (trunk: VLAN 10 + 20)
    eno1 ← tagged frames
    /   \
  eno1.10 eno1.20 ← tag removed:
  VLAN 10  VLAN 20 one network each
[dst MAC|src MAC|tag 10|data]
```

Term	Means
Tag	4 bytes (802.1Q) added to a frame: its VLAN ID, 1–4094
Access port	Switch port in one VLAN; frames leave it untagged
Trunk port	Carries many VLANs, every frame tagged
Native VLAN	The one VLAN a trunk sends without a tag

Making a VLAN interface (needs sudo; lost on reboot)

```
ip link add link eno1 name eno1.10 type vlan id 10 ← VLAN 10 on top of eno1
ip link add link main name main.10 type vlan id 10 ← same, if eno1 is a port of bridge main
ip link set eno1.10 up ← ip link del eno1.10 removes it
ip addr add 192.168.10.5/24 dev eno1.10 ← IP on the VLAN, not on eno1
ip -d link show eno1.10 ← check it; output reads:
6: eno1.10@eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 ... ← @eno1 = rides on eno1
    vlan protocol 802.1Q id 10 <REORDER_HDR> ... ← standard tagging, VLAN ID 10
```

VLAN-aware bridge (one bridge, many VLANs)

```
ip link set main type bridge vlan_filtering 1 ← switch on VLAN filtering
bridge vlan add dev eno1 vid 10 ← uplink: VLAN 10 allowed, tagged
bridge vlan add dev vnet0 vid 10 pvid untagged ← VM port: sees VLAN 10 untagged · check: bridge vlan show
```

Bonding modes

Mode	What it does	Switch setup
active-backup	One port works; the other takes over if it fails. Simplest	None
802.3ad (LACP)	All ports busy, split per connection, plus failover. Usual for speed	LACP group on the switch
balance-alb	Spreads traffic in and out across ports, plus failover	None
balance-rr	Packets take turns per port; can reorder TCP, so avoid	Static group

```
ip link add bond0 type bond mode active-backup miimon 100 ← check link every 100 ms
ip link set eno1 down; ip link set eno1 master bond0 ← a port must be down to join
ip link set eno2 down; ip link set eno2 master bond0; ip link set bond0 up
```

Reading cat /proc/net/bonding/bond0

```
Bonding Mode: fault-tolerance (active-backup)
Primary Slave: None
Currently Active Slave: eno1 ← port in use now
MII Status: up ← bond as a whole
MII Polling Interval (ms): 100 ← miimon
```

```
Slave Interface: eno1 ← one block per port
MII Status: up ← has link
Link Failure Count: 0
Slave Interface: eno2
MII Status: down ← cable/switch port dead
Link Failure Count: 3 ← times it dropped
```

Making it permanent with Netplan

```
vlan:
  eno1.10:
    id: 10
    link: eno1 ← main if eno1 is bridged
    addresses: [192.168.10.5/24]
```

```
bonds:
  bond0:
    interfaces: [eno1, eno2]
    parameters:
      mode: active-backup
      mii-monitor-interval: 100
```

Put both under **network:** (beside **ethernets:**). Bridge on a bond: **interfaces:** **[bond0]** under **main**.

Remember: the switch must match: a **trunk** port for VLANs, an **LACP group** for **802.3ad**.